

Gurkirat Panesar

GIS Analyst

- GIS Portfolio
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EDUCATION

Bachelor of Environmental Studies Honours Geomatics Co-op

University of Waterloo
Expected October 2026

Coursework: Advanced GIS, Spatial Databases, Geodesy and Surveying, Advanced Remote Sensing Techniques, Geospatial Data Science, Designing Effective Maps, Remotely Piloted Aircraft Systems, Multivariate Statistics, Data Types & Structures

TECHNICAL SKILLS

GIS & Web GIS

ArcGIS Pro, ArcGIS Online, Experience Builder, StoryMaps, ArcGIS Hub, QGIS, Google Earth Engine, ArcUrban, AutoCAD, Survey123

GIS Analysis & Automation

ArcPy, ModelBuilder, Geocoding, Spatial Joins, Network Datasets, Georeferencing, Cartographic Design, QA/QC

Spatial Databases & Data

SQL, PostgreSQL, MS SQL Server, MySQL, Excel, Power BI

Programming

Python, JavaScript, TypeScript, R

Surveying & Field Methods

Total Station, Trimble GNSS/GPS Receivers, RPAS/UAV Surveying, Traversing, Levelling, Hydrometric Surveying

CERTIFICATIONS

Esri ArcGIS Pro Associate 2025

In Progress; Expected 2026

FME Desktop

May 2026

GeoAI in ArcGIS Pro and Survey123

May 2026

SUMMARY

Geomatics graduate with experience in municipal GIS, spatial analysis, web mapping, automation, and spatial data management. Skilled in ArcGIS, Python, JavaScript, SQL, and geospatial QA/QC, with a strong ability to improve workflows and deliver clear, reliable spatial solutions.

EXPERIENCE

GIS Technician

Region of Peel

May 2025 - Aug 2025
Brampton, ON

- Automated and supported municipal building permit GIS workflows using ArcMap ModelBuilder and ArcGIS Pro with Python, resolving field-mapping, truncation, and geocoding logic issues in large permit datasets.
- Improved geocoding reliability by refining address input logic, locator configuration, match thresholds, and rematch workflows across municipal datasets.
- Developed a reusable ArcGIS Pro Python tool to update Archaeological Sites data in PeelDB, improving repeatability and reducing manual processing.
- Converted external API JSON data into GIS feature classes, performed ward-based spatial joins, and produced filtered, client-ready outputs.
- Maintained enterprise GIS metadata and attribute domains, performed QA/QC, and documented workflows for long-term handover.

GIS Associate

University of Waterloo - Geospatial Centre

Sep 2024 - Dec 2024
Waterloo, ON

- Built an Air Photo Collection Dashboard in ArcGIS Experience Builder, enabling users to search and download thousands of air photos by year and municipality.
- Developed a University of Waterloo campus paths network dataset from OpenStreetMap data and created a 20-minute Campus Green Space Tour map using ArcGIS StoryMaps.
- Created an ArcGIS Hub open data portal to streamline access to air photos and spatial datasets from the ArcServer repository.
- Processed and updated 3 million Teranet property records in PostgreSQL, linked records to property boundary shapefiles, and clipped outputs by municipality using ArcPy.
- Maintained and organized thousands of open GIS datasets while supporting users with data discovery, air photo access, and GIS troubleshooting.

Data Analytics Technician

The Regional Municipality of York - Corporate Services

Jan 2024 - Apr 2024
Newmarket, ON

- Conducted GIS suitability analysis for affordable housing redevelopment on provincial lands by evaluating floodplains, highways, protected areas, existing parcels, and other spatial constraints.
- Researched zoning bylaw policies and developed an interactive Yellow Belt map for York Region municipalities using ArcGIS Pro and ArcGIS Online.
- Created maps and layouts showing housing supply and development statistics for York Region's 2023 Official Plan.
- Supported migration from the MS Access Regional Street Naming database by applying Agile user story standards and improving data management workflows.
- Maintained MS SQL Server spatial data, troubleshoot errors, and improved database quality and consistency.

PROJECTS

GeoDetect AI ([Link](#))

React | Vite | Transformers.js | Geospatial AI

- Developed a browser-based remote sensing AI tool that detects user-defined features in satellite imagery through custom text prompts, with interactive map-based area selection and imagery upload support.
- Implemented a detection review and export pipeline with GeoTIFF metadata extraction, georeferencing, tuneable tiled scanning, spectral refinement, and exports to standard spatial data formats.

Sentinel-2 Snow Explorer and Analyzer ([Link](#))

Remote Sensing Capstone | Google Earth Engine | Sentinel-2 | JavaScript

- Built an interactive Google Earth Engine application to compare urban-rural snow persistence across Thunder Bay, Sudbury, and Sault Ste. Marie using Sentinel-2 imagery.
- Integrated NDSI snow masking, fractional snow cover, adaptive thresholding, change detection, ESA WorldCover land-cover analysis, terrain variables, shoreline and road proximity, and site-verification workflows.

University of Waterloo Campus Map Redesign ([Link](#))

GIS Capstone | ArcGIS Pro | ArcGIS Online | Experience Builder

- Designed an ArcGIS Online and Experience Builder campus wayfinding app with searchable buildings, amenities, accessibility layers, pedestrian routing, pop-ups, and responsive desktop and mobile layouts.
- Cleaned, validated, and organized campus spatial datasets to improve map legibility, visual hierarchy, layer filtering, and navigation across the University of Waterloo main campus.